THE STATE UNIVERSITY OF ZANZIBAR

SCHOOL OF BUSINESS

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

CASHFLOW STSTEM– ANDROID APPLICATION

Project Report

Programme : Bachelor of Information Technology with Accounting (BITA) Institution:

State University of Zanzibar (SUZA)

Supervisor: Mr. Massoud Hamad

Year: 2026

Group Members

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INTRODUCTION

The Cash Flow Management System is a web-based application designed to help businesses manage their daily financial transactions efficiently. The system records cash inflows and cash outflows, calculates the available cash balance automatically, and generates financial reports. It replaces manual bookkeeping methods with a computerized solution that improves accuracy, transparency, and decision-making.

PROBLEM STATEMENT

Many small and medium businesses still record their daily income and expenses manually using notebooks or spreadsheets. This creates several challenges including:

1. Loss of financial records.
2. Inaccurate cash calculations.
3. Difficulty in tracking daily transactions.
4. Time-consuming report preparation.
5. Poor financial decision making.
6. Lack of secure data storage.

PROPOSED SOLUTION

Android-Based Cashflow Management Application

To solve these financial management issues, we developed an Android application that enables:

1. Digital Registration of Income Sources and Expense Categories
   Registration of various income categories (e.g., Sales, Investments, Loans, Grants).
   Registration of distinct expense categories (e.g., Rent, Salaries, Inventory, Utilities, Marketing).
2. Real-Time Cash In and Cash Out Tracking
   Instant recording and monitoring of money received (Cash In) and money spent (Cash Out).
   Real-time tracking of the available cash balance to ensure healthy liquidity and prevent overspending.
3. Financial Transaction CRUD Operations
   Full capability to Create, Read, Update, and Delete transaction records, making it easy to correct bookkeeping errors or update financial history.
4. Admin Login and Cashflow Dashboard
   A secure login portal for authorized business owners, administrators, or managers.
   An intuitive dashboard providing visual insights (such as graphs or pie charts) of income vs. expenses, net profit margins, and daily/monthly trends.
5. User-Friendly and Accessible Financial System for Managers
   A clean, non-technical interface designed for users without an accounting background to effortlessly manage budgets and generate instant financial summaries.

PROJECT OBJECTIVES

1. To develop a mobile-friendly platform for real-time cashflow data management.
2. To enable seamless recording, tracking, and updating of financial transactions (Income and Expenses).
3. To minimize paperwork, manual bookkeeping errors, and financial discrepancies.
4. To improve financial data accessibility and instant report generation through smartphone use.

FUNCTIONAL

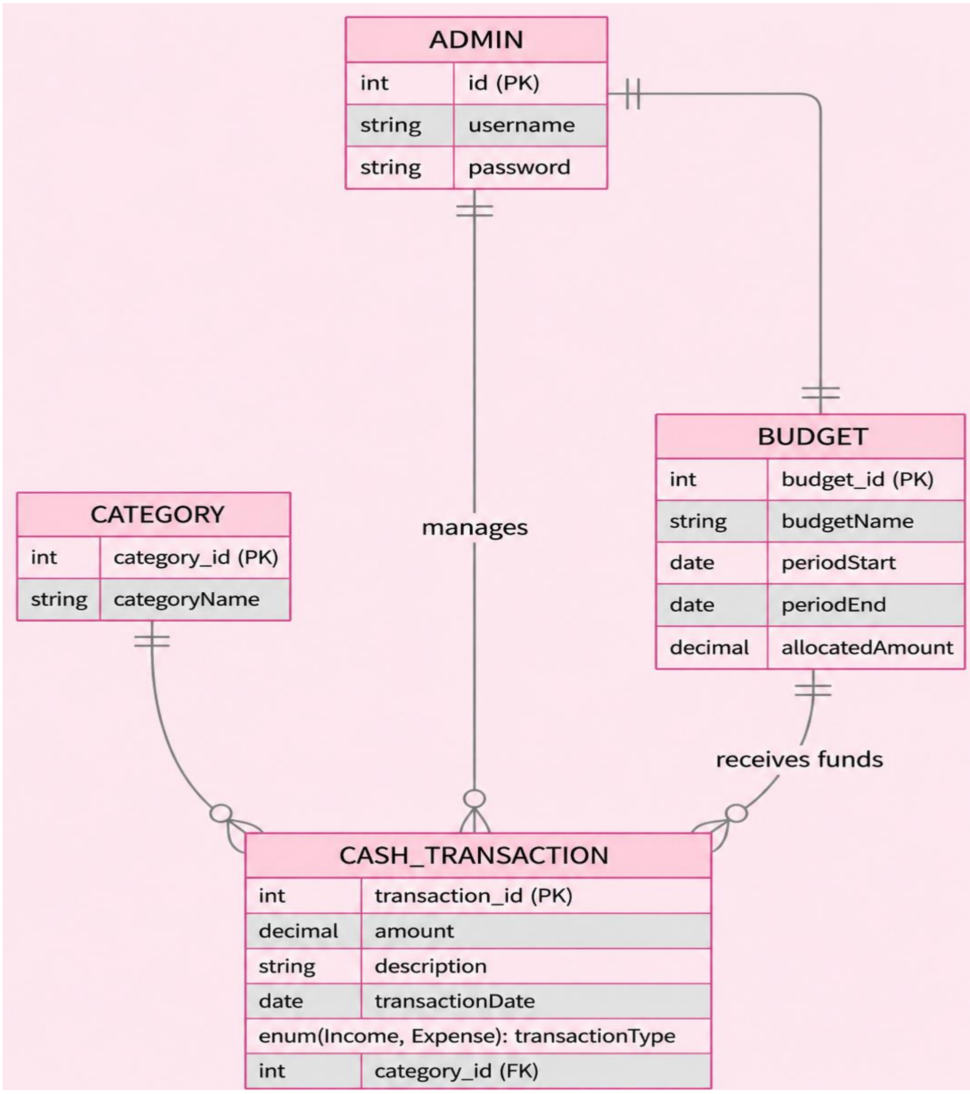
1. Admin login with authentication
2. Dashboard displaying counts of data
3. Add/Edit/Delete user records
4. Add/Edit/Delete guardian records
5. Add/Edit/Delete count cashflow

6. Sidebar navigation menu

NON-FUNCTIONAL REQUIREMENTS

- Responsiveness: The app must be fully responsive, optimizing UI layouts to ensure seamless performance and visual consistency across a wide range of Android devices and screen sizes.
- Security: Financial data and user login credentials must be stored securely using strong encryption protocols, secure session management, and protected local storage to prevent unauthorized access.
- Performance: The system must ensure high-speed data fetching and smooth list scrolling by utilizing Recycler View and efficient data caching for thousands of financial transaction logs.
- Scalability: The app's architecture must be decoupled and scalable, allowing it to be easily extended to a web version or integrated with cloud databases to accommodate growing transaction volumes.
- Maintainability: The codebase must be highly modular and clean, utilizing Retrofit for reliable API communication and structured under the MVC (Model-View-Controller) architecture to simplify future updates and debugging.

ENTITY RELATIONSHIP DIAGRAM (ERD)


ADMIN
int
id (PK)
string
username
string
password
BUDGET
int
budget_id (PK)
string
budgetName
date
periodStart
date
periodEnd
decimal
allocatedAmount
CATEGORY
int
category_id (PK)
string
categoryName
manages
receives funds
CASH_TRANSACTION
int
transaction_id (PK)
decimal
amount
string
description
date
transactionDate
enum(Income, Expense): transactionType
int
category_id (FK)

TECHNOLOGIES USED

1. Kotlin: Android development language
2. Android Studio: IDE for app development
3. SpringBoot:IntelJ
4. Retrofit: API client for server connection

BACKEND API IMPLEMENTATION

```
@PostMapping
public ResponseEntity<ApiResponse> saveOrUpdate(
        @RequestHeader("X-User-Id") Long userId,
        @Valid @RequestBody EntryDto dto) {
    ApiResponse response = transactionService.saveOrUpdateCategory(userId, dto);
    return ResponseEntity.status(response.isSuccess() ? HttpStatus.OK : HttpStatus.BAD_RE
            .body(response);
}

@GetMapping("/{id}")
public ResponseEntity<ApiResponse> getById(@PathVariable Long id) {
    ApiResponse response = transactionService.getTransactionById(id);
    return ResponseEntity.status(response.isSuccess() ? HttpStatus.OK : HttpStatus.NOT_FO
            .body(response);
}

@GetMapping("/user")
public ResponseEntity<ApiResponse> getTransactionsForUser(
        @RequestHeader("X-User-Id") Long userId) {
    ApiResponse response = transactionService.getTransactionByUserId(userId);
    return ResponseEntity.ok(response);
}
```

```java
@GetMapping("/user/type")
public ResponseEntity<ApiResponse> getTransactionsByUserAndType(
        @RequestHeader("X-User-Id") Long userId,
        @RequestParam TransactionType type) {
    ApiResponse response = transactionService.getByUserIdAndType(userId, type);
    return ResponseEntity.ok(response);
}

@GetMapping("/all")
public ResponseEntity<ApiResponse> getAll() {
    ApiResponse response = transactionService.getAll();
    return ResponseEntity.ok(response);
}

@GetMapping("/summary")
public ResponseEntity<ApiResponse> getTransactionSummary(
        @RequestParam(required = false) Long userId,
        @RequestParam(required = false) String name,
        @RequestParam(required = false) Long id,
        @RequestParam(required = false) String dateRange,
        @RequestParam(required = false) @DateTimeFormat(iso = DateTimeFormat.ISO.DATE_TIME)
        @RequestParam(required = false) @DateTimeFormat(iso = DateTimeFormat.ISO.DATE_TIME)

    ApiResponse response = transactionService.getTransactionSummary(
            userId, name, id, dateRange, startDate, endDate);
    return ResponseEntity.ok(response);
}
```

C:\Windows\System32\cmd.exe - mvnw spring-boot:run

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\lady_programmer\Desktop\cash_flow\cash_flow\backend>mvnw spring-boot:run
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.thuwaiba.cashFlow:cashFlow >-------------------
[INFO] Building  0.0.1-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] >>> spring-boot:4.1.0:run (default-cli) > test-compile @ cashFlow >>>
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ cashFlow ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ cashFlow ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ cashFlow ---
[INFO] skip non existing resourceDirectory C:\Users\lady_programmer\Desktop\cash_flow\cash_flow\backend\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ cashFlow ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] <<< spring-boot:4.1.0:run (default-cli) < test-compile @ cashFlow <<<
[INFO]
[INFO]
[INFO] --- spring-boot:4.1.0:run (default-cli) @ cashFlow ---
[INFO] Attaching agents: []

  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
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 :: Spring Boot ::                (v4.1.0)
```

CRUD BUTTON IMPLEMENTATION

Each item in the list (Cash Transaction / Category / Budget) includes two icon buttons:

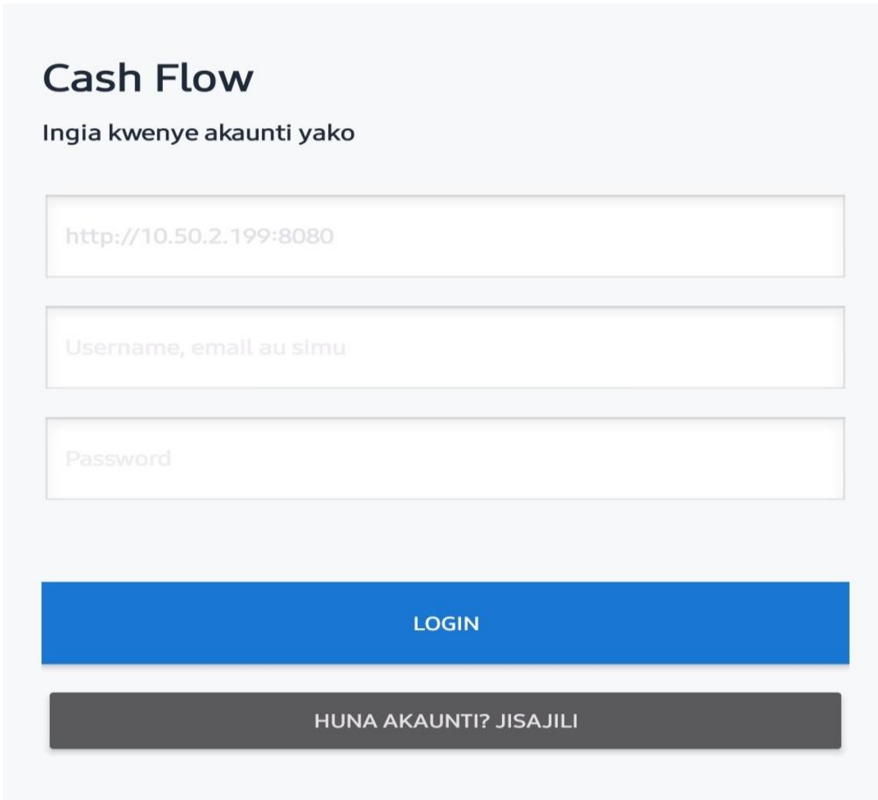
- Edit: Opens an AlertDialog pre-filled with the existing financial transaction or category data for updating.
- Delete: Triggers a backend API delete call to remove the entry from the cashflow records.
- Status: Triggers a backend API status call status the entry from the cashflow records

These are connected via onClickListeners in the Adapter class:

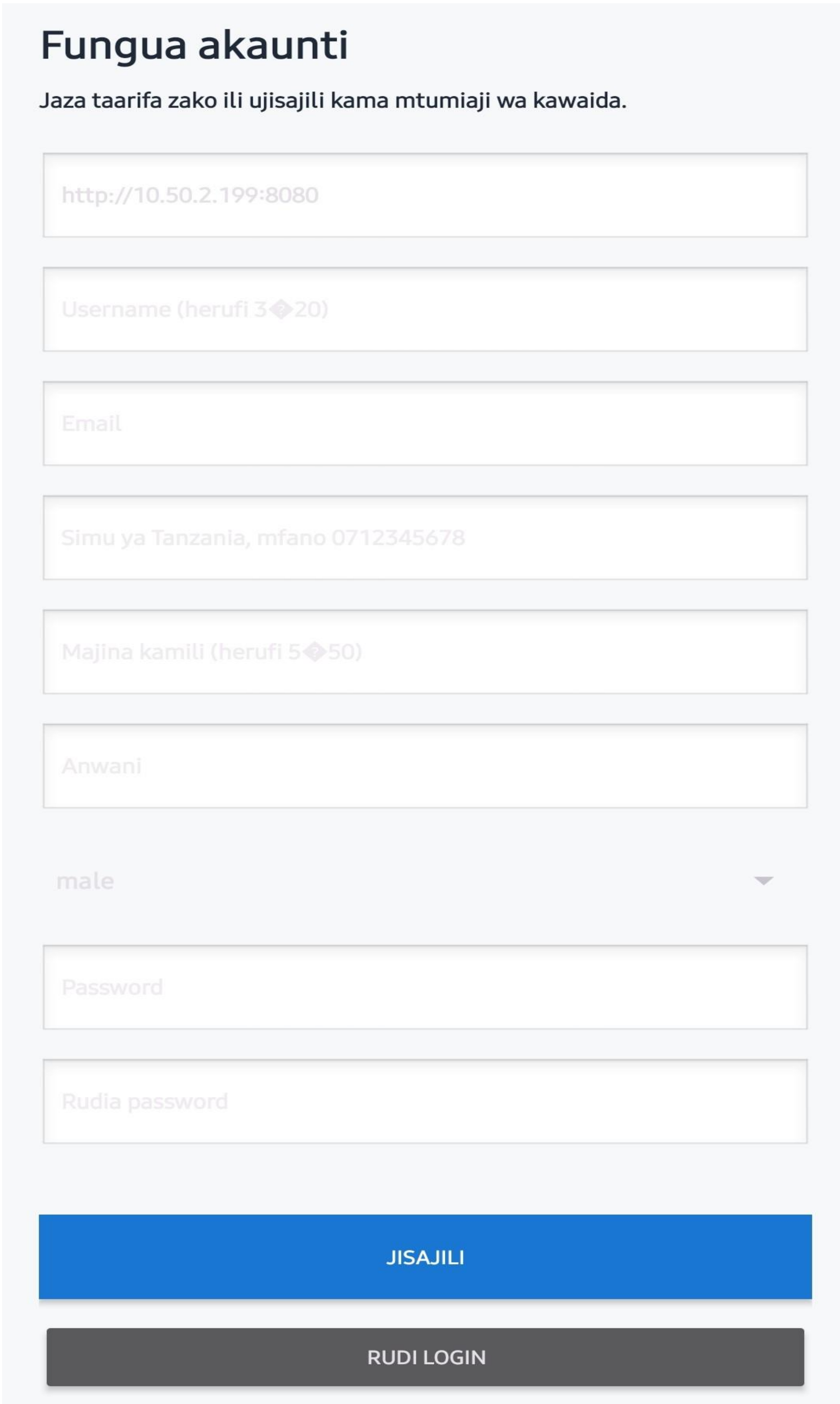
```
btnEdit.setOnClickListener { onEdit(item) }
btnDelete.setOnClickListener { onDelete(item) }
btnStatus.setOnClickListener { onStatus(item) }
```

UI DESIGNS:

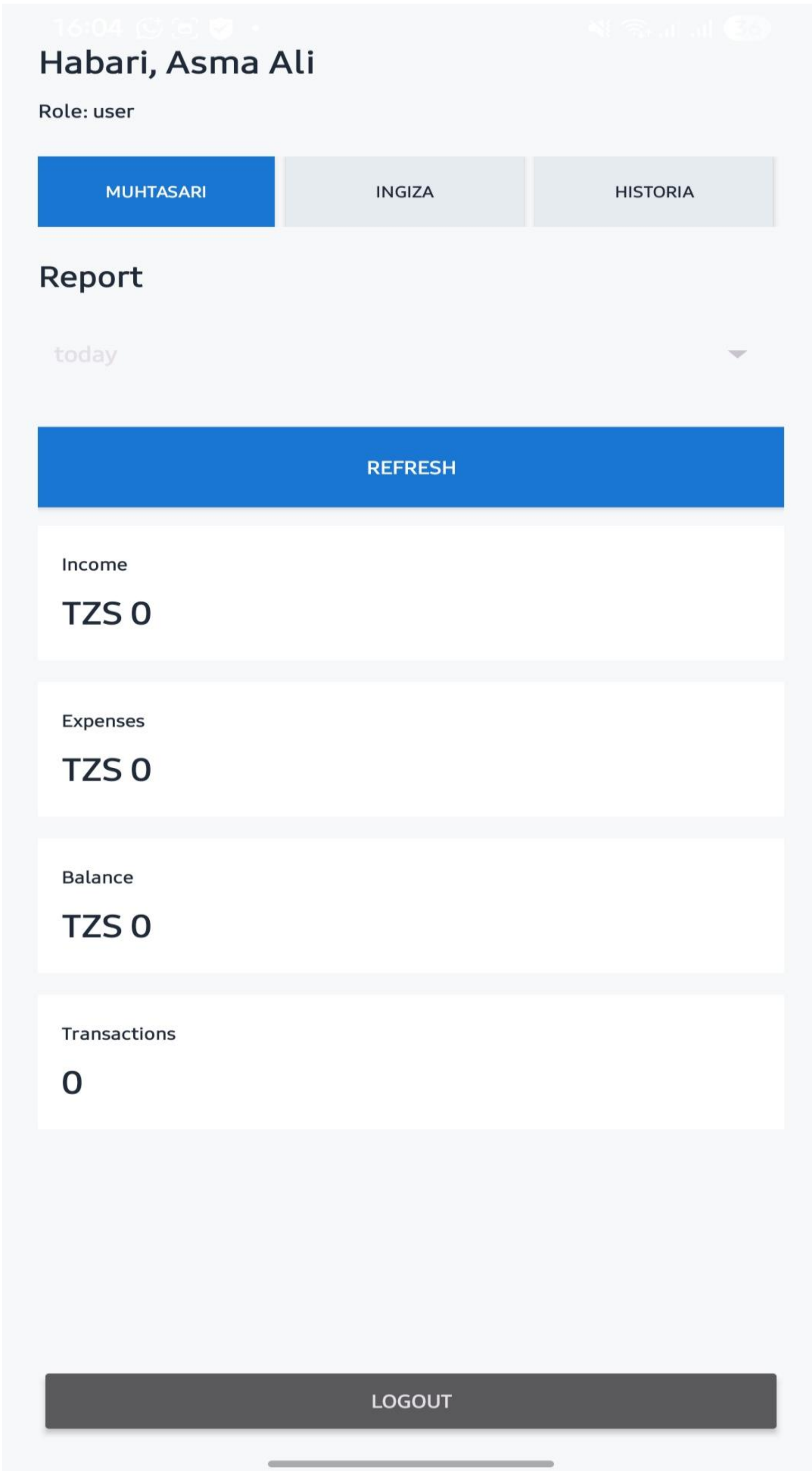
Login users

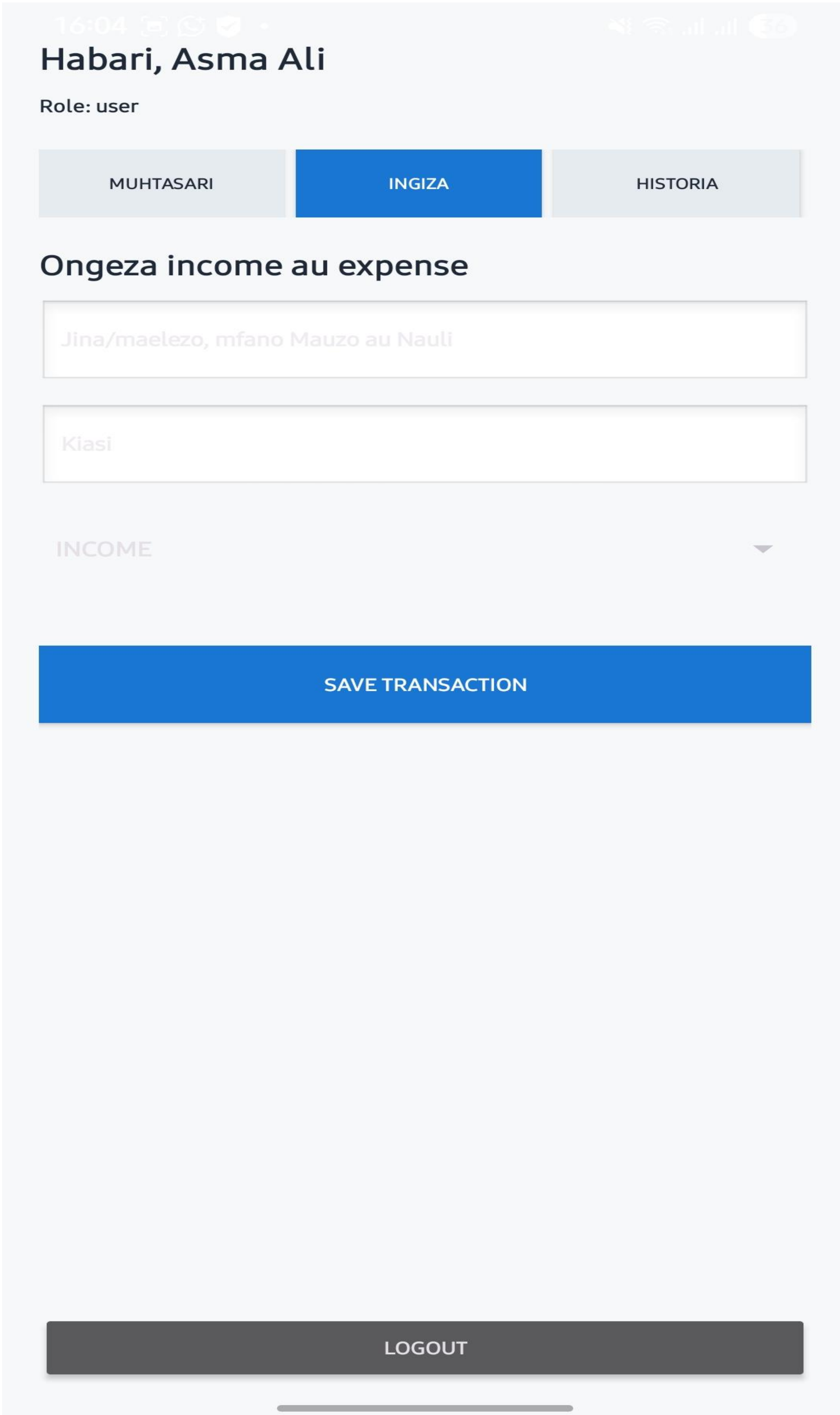


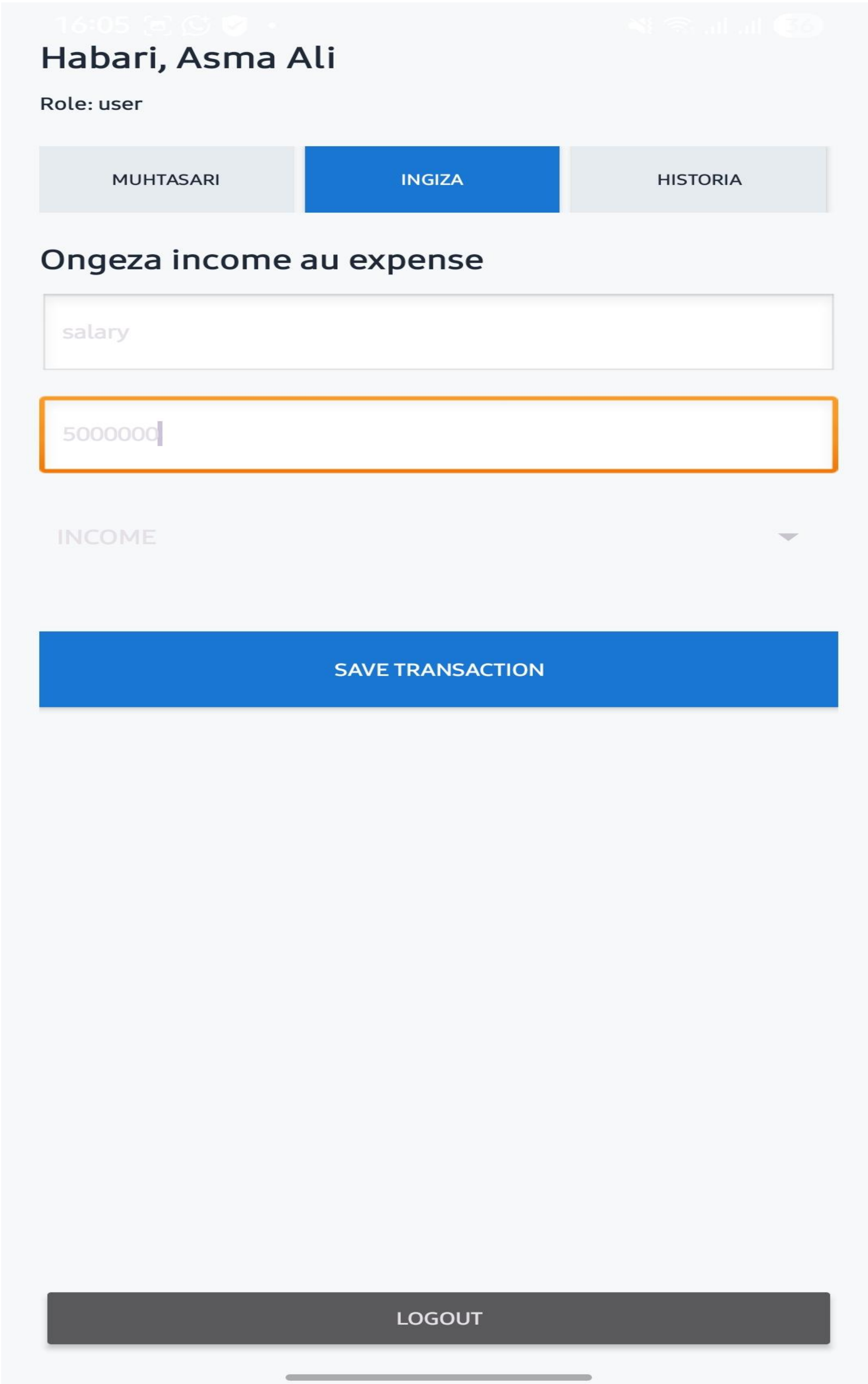
Register users:

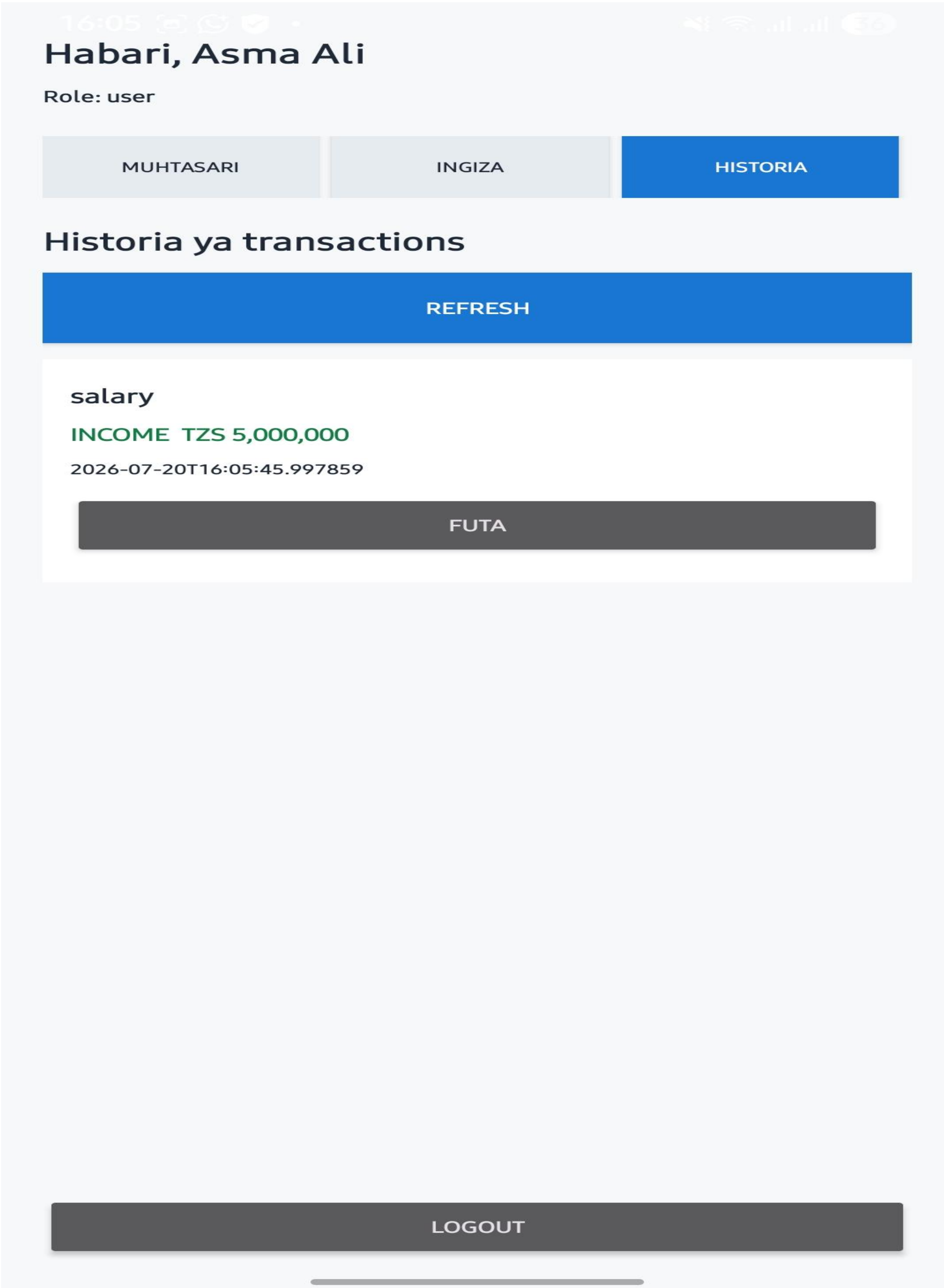


User dashboard:




Habari, Asma Ali
Role: user
MUHTASARI
INGIZA
HISTORIA
Ongeza income au expense
Jina/maelezo, mfano Mauzo au Nauli
Kiasi
INCOME
SAVE TRANSACTION
LOGOUT


Habari, Asma Ali
Role: user
MUHTASARI
INGIZA
HISTORIA
Ongeza income au expense
salary
5000000
INCOME
SAVE TRANSACTION
LOGOUT


Habari, Asma Ali
Role: user
MUHTASARI
INGIZA
HISTORIA
Historia ya transactions
REFRESH
salary
INCOME TZS 5,000,000
2026-07-20T16:05:45.997859
FUTA
LOGOUT

16:09

Habari, Asma Ali

Role: user

MUHTASARI | INGIZA | HISTORIA

Historia ya transactions

REFRESH

salary

INCOME TZS 5,000,000

2026-07-20T16:05:45.997859

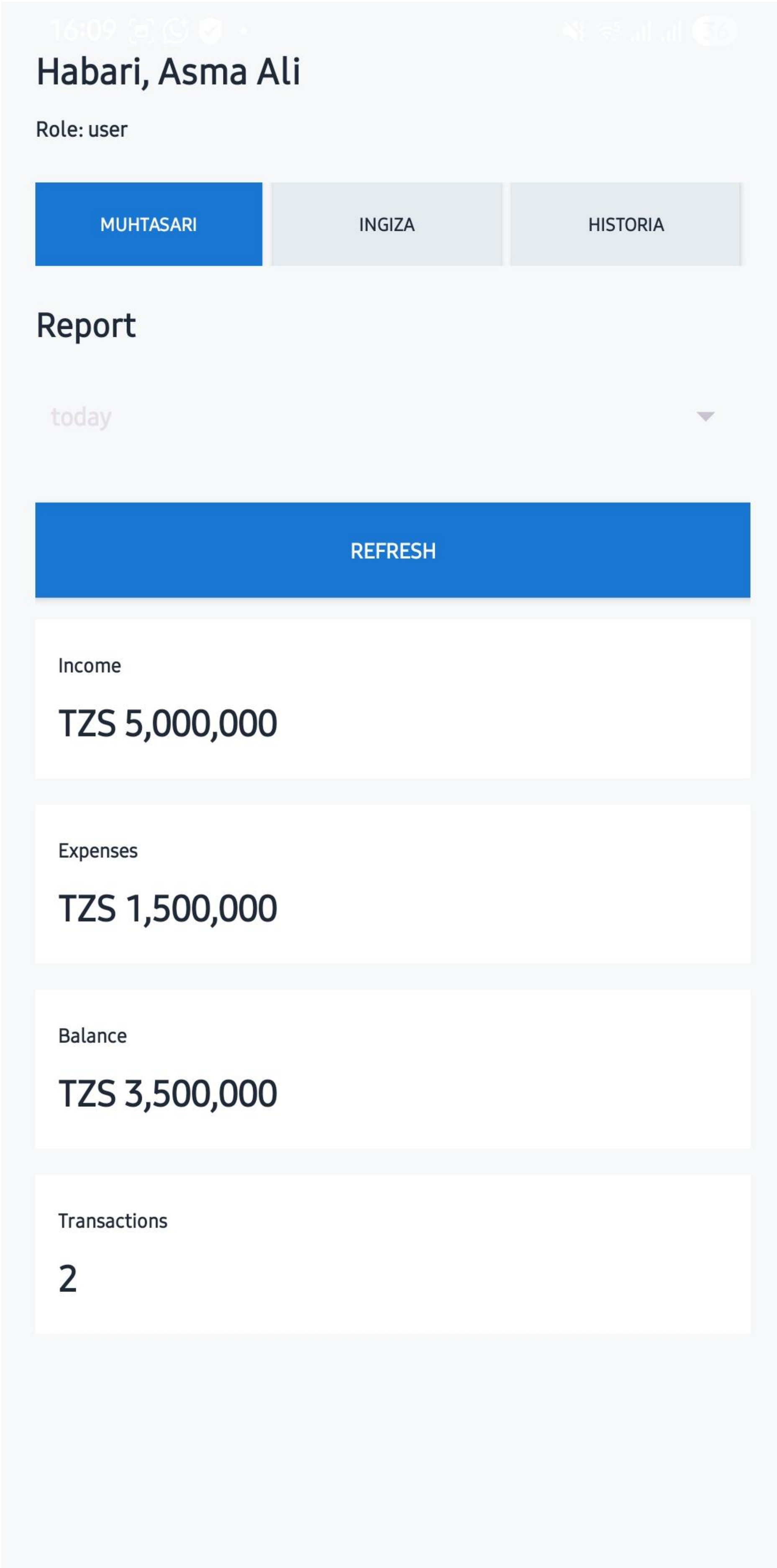
FUTA

hostel

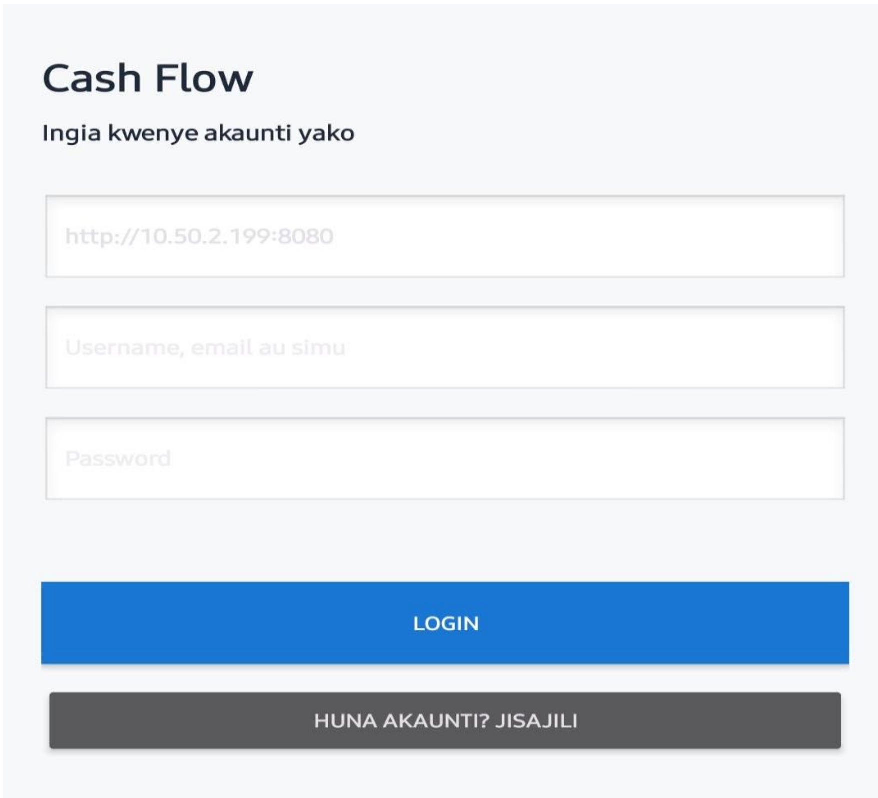
EXPENSE TZS 1,500,000

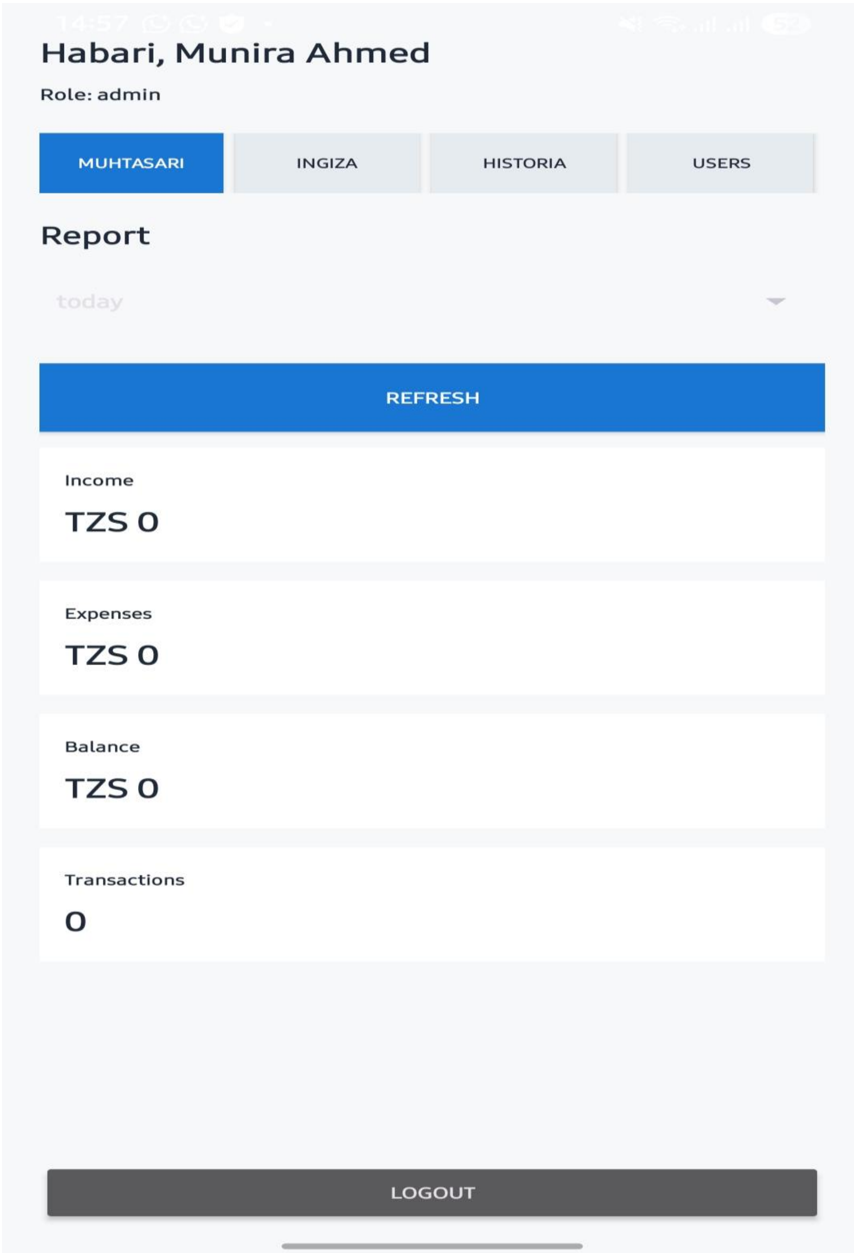
2026-07-20T16:09:06.542944

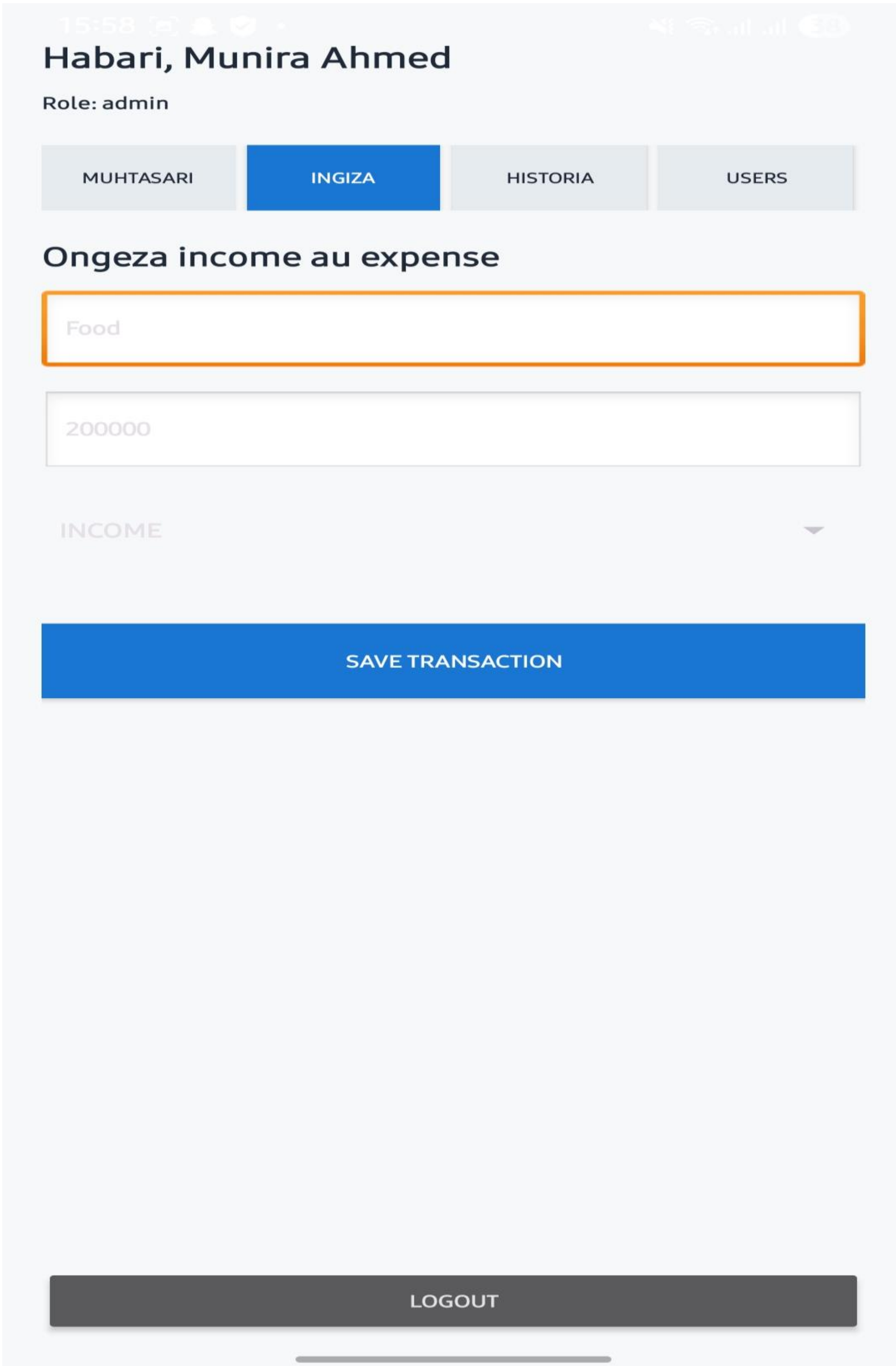
FUTA

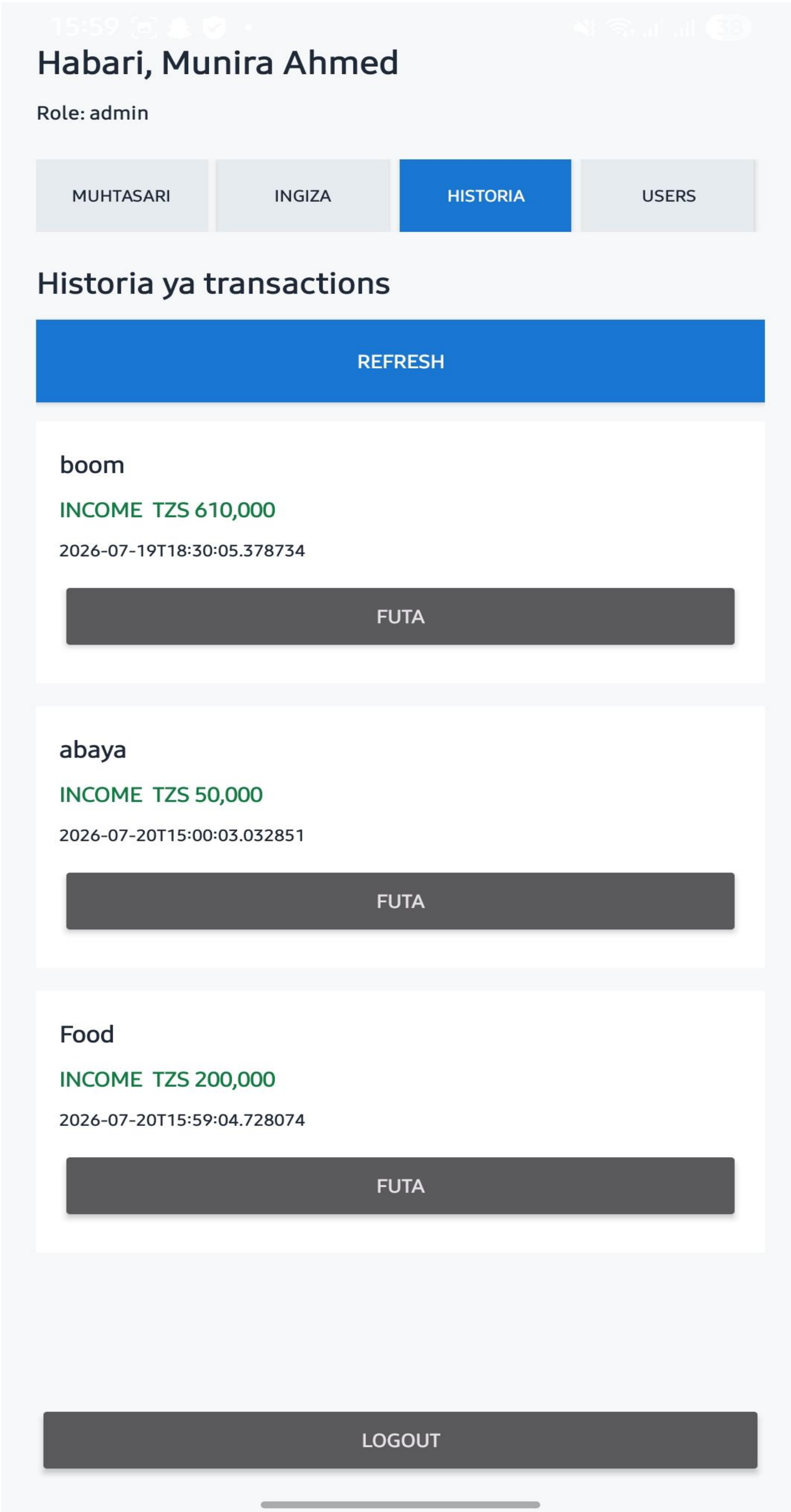

16:09
Habari, Asma Ali
Role: user
MUHTASARI
INGIZA
HISTORIA
Report
today
REFRESH
Income
TZS 5,000,000
Expenses
TZS 1,500,000
Balance
TZS 3,500,000
Transactions
2

ADMIN:

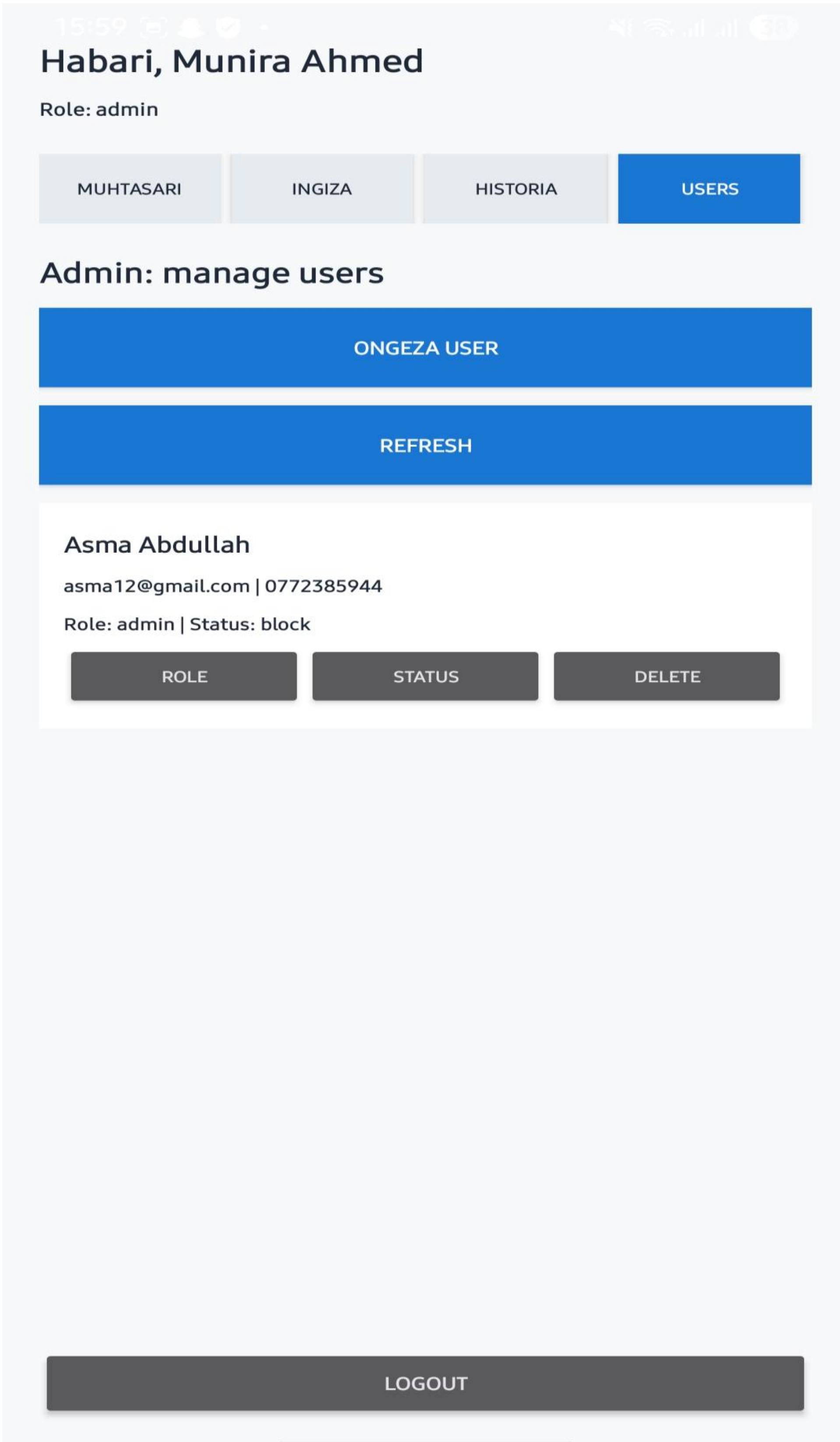

Cash Flow
Ingia kwenye akaunti yako
http://10.50.2.199:8080
Username, email au simu
Password
LOGIN
HUNA AKAUNTI? JISAJILI


Habari, Munira Ahmed
Role: admin
MUHTASARI
INGIZA
HISTORIA
USERS
Report
today
REFRESH
Income
TZS 0
Expenses
TZS 0
Balance
TZS 0
Transactions
0
LOGOUT


Habari, Munira Ahmed
Role: admin
MUHTASARI
INGIZA
HISTORIA
USERS
Ongeza income au expense
Food
200000
INCOME
SAVE TRANSACTION
LOGOUT


Habari, Munira Ahmed
Role: admin
MUHTASARI
INGIZA
HISTORIA
USERS
Historia ya transactions
REFRESH
boom
INCOME TZS 610,000
2026-07-19T18:30:05.378734
FUTA
abaya
INCOME TZS 50,000
2026-07-20T15:00:03.032851
FUTA
Food
INCOME TZS 200,000
2026-07-20T15:59:04.728074
FUTA
LOGOUT


16:00
38
Habari, Munira Ahmed
Role: admin
MUHTASARI
INGIZA
HISTORIA
USERS
Ongeza User
Username
Email
Phone
Full name
Address
Password
female
user
Cancel
Save
LOGOUT


Habari, Munira Ahmed
Role: admin
MUHTASARI
INGIZA
HISTORIA
USERS
Admin: manage users
ONGEZA USER
REFRESH
Asma Abdullah
asma12@gmail.com | 0772385944
Role: admin | Status: block
ROLE
STATUS
DELETE
LOGOUT

CONCLUSION

This project effectively addresses the needs of modern liquidity and finance tracking by adopting a digital approach to cashflow management. It significantly improves data accessibility, recording accuracy, and real-time balance updates while minimizing paperwork and human bookkeeping errors. Furthermore, the decoupling of its architecture ensures that the application can be seamlessly extended into a comprehensive, multi-platform financial information system with both web and distributed database support.